

Mechanics

Third Edition

Course of Theoretical physics

Volume 1.

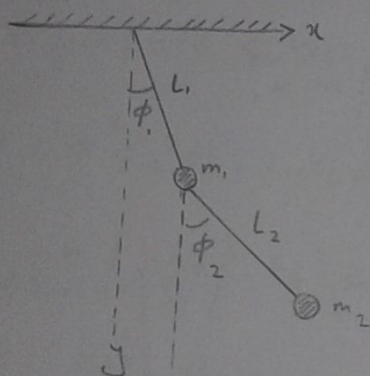
L. D. Landau & E. M. Lifshitz

Solution Manual (written by Nader Safavinia)

Problem 1

Page 11

* It's almost solved in the book, but for the sake of Comprehensibility:



$$\text{for } m_1 \quad \begin{cases} T_1 = \frac{1}{2} m_1 L_1^2 \dot{\phi}_1^2 \\ U_1 = -m_1 g L_1 \cos \phi_1 \end{cases} \Rightarrow \mathcal{L}_1 = T_1 - U_1$$

$$\Rightarrow \mathcal{L}_1 = \frac{1}{2} m_1 L_1^2 \dot{\phi}_1^2 + m_1 g L_1 \cos \phi_1$$

$$\text{for } m_2 \quad \begin{cases} T_2 = \frac{1}{2} m_2 (\dot{x}^2 + \dot{y}^2) \\ U_2 = -m_2 g (L_2 \cos \phi_2 + L_1 \cos \phi_1) \end{cases} : \begin{cases} x = L_1 \sin \phi_1 + L_2 \sin \phi_2 \\ \dot{x} = L_1 \dot{\phi}_1 \cos \phi_1 + L_2 \dot{\phi}_2 \cos \phi_2 \\ y = L_1 \cos \phi_1 + L_2 \cos \phi_2 \\ \dot{y} = -L_1 \dot{\phi}_1 \sin \phi_1 - L_2 \dot{\phi}_2 \sin \phi_2 \end{cases}$$

$$T_2 = \frac{1}{2} m_2 (\dot{x}^2 + \dot{y}^2) = \frac{1}{2} m_2 \left[(L_1 \dot{\phi}_1 \cos \phi_1 + L_2 \dot{\phi}_2 \cos \phi_2)^2 + (-L_1 \dot{\phi}_1 \sin \phi_1 - L_2 \dot{\phi}_2 \sin \phi_2)^2 \right]$$

$$= \frac{1}{2} m_2 \left[L_1^2 \dot{\phi}_1^2 \cos^2 \phi_1 + L_2^2 \dot{\phi}_2^2 \cos^2 \phi_2 + 2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos \phi_1 \cos \phi_2 + L_1^2 \dot{\phi}_1^2 \sin^2 \phi_1 + L_2^2 \dot{\phi}_2^2 \sin^2 \phi_2 + 2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \sin \phi_1 \sin \phi_2 \right]$$

$$T_2 = \frac{1}{2} m_2 (\dot{L}_1^2 \dot{\phi}_1^2 + \dot{L}_2^2 \dot{\phi}_2^2 + 2 \dot{L}_1 \dot{L}_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2))$$

$$\mathcal{L}_2 = T_2 - U_2 = \frac{1}{2} m_2 \dot{L}_1^2 \dot{\phi}_1^2 + \frac{1}{2} m_2 \dot{L}_2^2 \dot{\phi}_2^2 + m_2 \dot{L}_1 \dot{L}_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2) +$$

$$\hookrightarrow m_2 g L_2 \cos \phi_2 + m_2 g L_1 \cos \phi_1$$

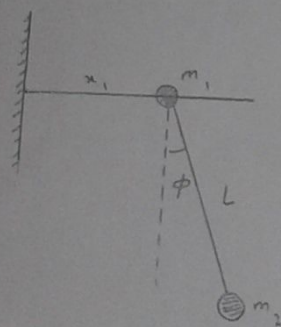
$$\mathcal{L}_{tot} = \mathcal{L}_1 + \mathcal{L}_2 = \frac{1}{2} (m_1 + m_2) \dot{L}_1^2 \dot{\phi}_1^2 + \frac{1}{2} m_2 \dot{L}_2^2 \dot{\phi}_2^2 + m_2 \dot{L}_1 \dot{L}_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2) +$$

$$\hookrightarrow (m_1 + m_2) g L_1 \cos \phi_1 + m_2 g L_2 \cos \phi_2$$

Problem 2

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1. Considering the same reason as * :



$$\text{for } \underline{m_1} \quad \begin{cases} T_1 = \frac{1}{2} m_1 \dot{x}_1^2 \\ U_1 = 0 \end{cases}$$

$$\text{for } \underline{m_2} \quad \begin{cases} T_2 = \frac{1}{2} m_2 (\dot{x}^2 + \dot{y}^2) \\ U_2 = -m_2 g L \cos \phi \end{cases} \quad \begin{cases} x = x_1 + L \sin \phi \\ \dot{x} = \dot{x}_1 + L \dot{\phi} \cos \phi \\ y = L \cos \phi \\ \dot{y} = -L \dot{\phi} \sin \phi \end{cases}$$

$$\mathcal{L}_{tot} = \mathcal{L}_1 + \mathcal{L}_2 = T_1 - U_1 + T_2 - U_2 = \frac{1}{2} m_1 \dot{x}_1^2 - 0 + \frac{1}{2} m_2 [(\dot{x}_1 + L \dot{\phi} \cos \phi)^2 +$$

$$\hookrightarrow (-L \dot{\phi} \sin \phi)^2] + m_2 g L \cos \phi$$

$$\mathcal{L}_{tot} = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_1^2 + \frac{1}{2} m_2 (L^2 \dot{\phi}^2 \cos^2 \phi + 2 \dot{x}_1 L \dot{\phi} \cos \phi + L^2 \dot{\phi}^2 \sin^2 \phi)$$

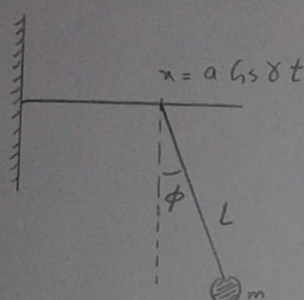
$$\hookrightarrow + m_2 g L \cos \phi$$

$$\text{III} : m a \delta L \dot{\phi} \sin(\phi - \delta t) = - \frac{d}{dt} (m L a \delta \cos(\phi - \delta t)) + m L a \delta^2 \sin(\phi - \delta t)$$

this term can be omitted

$$\mathcal{L} = \frac{1}{2} m L^2 \dot{\phi}^2 + m L a \delta^2 \sin(\phi - \delta t) + m g L \cos \phi$$

b



for the object

$$\begin{cases} x = a \cos \delta t + L \sin \phi \\ \dot{x} = -a \delta \sin \delta t + L \dot{\phi} \cos \phi \\ y = L \cos \phi \\ \dot{y} = -L \dot{\phi} \sin \phi \end{cases}$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) = \frac{1}{2} m [(-a \delta \sin \delta t + L \dot{\phi} \cos \phi)^2 + (-L \dot{\phi} \sin \phi)^2]$$

$$T = \frac{1}{2} m (a^2 \delta^2 \sin^2 \delta t + L^2 \dot{\phi}^2 \cos^2 \phi - 2 a \delta L \dot{\phi} \sin \delta t \cos \phi + L^2 \dot{\phi}^2 \sin^2 \phi)$$

$$U = -m g L \cos \phi$$

$$\mathcal{L} = T - U = \frac{1}{2} m L^2 \dot{\phi}^2 \cos^2 \phi + \frac{1}{2} m L^2 \dot{\phi}^2 \sin^2 \phi + \frac{1}{2} m a^2 \delta^2 \sin^2 \delta t - m a \delta L \dot{\phi} \sin \delta t \cos \phi + m g L \cos \phi$$

I

II

$$\text{I} : \sin^2 \delta t = \frac{1}{2} (1 - \cos 2\delta t) \Rightarrow \frac{1}{2} m a^2 \delta^2 \sin^2 \delta t = \frac{1}{4} m a^2 \delta^2 \frac{d}{dt} \left(t - \frac{1}{2\delta} \sin 2\delta t \right)$$

This term can be omitted

II

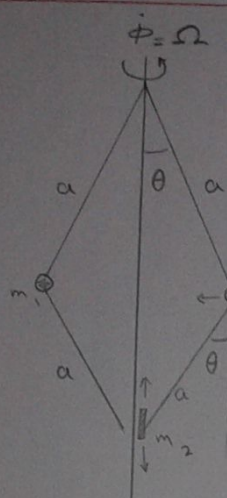
$$-m a \delta L \dot{\phi} \sin \delta t \cos \phi = - \frac{d}{dt} (m a \delta L \sin \delta t \sin \phi) + m a L \delta^2 \cos \delta t \sin \phi$$

This term can be omitted, too

$$\mathcal{L} = \frac{1}{2} m L^2 \dot{\phi}^2 + m a L \delta^2 \cos \delta t \sin \phi + m g L \cos \phi$$

Problem 4

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$$\begin{cases} dl_1 = a d\theta + a \sin \theta d\phi \\ dl_1^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2 \\ dl_2 = -2a \sin \theta d\theta \quad (L_2 = 2a \cos \theta) \\ dl_2^2 = 4a^2 \sin^2 \theta d\theta^2 \end{cases}$$

$$T = 2 \times \frac{1}{2} m_1 (a^2 \dot{\theta}^2 + a^2 \sin^2 \theta \dot{\phi}^2) + \frac{1}{2} m_2 (4a^2 \sin^2 \theta \dot{\theta}^2)$$

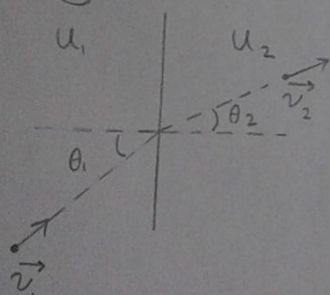
$$U = -2m_1 g a \cos \theta - m_2 g (2a \cos \theta)$$

$$L = T - U = m_1 (a^2 \dot{\theta}^2 + a^2 \sin^2 \theta \Omega^2) + 2m_2 a^2 \sin^2 \theta \dot{\theta}^2 + 2(m_1 + m_2) g a \cos \theta$$

$$L = m_1 a^2 (\dot{\theta}^2 + \Omega^2 \sin^2 \theta) + 2m_2 a^2 \dot{\theta}^2 \sin^2 \theta + 2(m_1 + m_2) g a \cos \theta$$

Problem

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$$E_{\text{before}} = E_{\text{after}}$$

$$(T + U_1)_b = (T + U_2)_a$$

$$\frac{1}{2} m |\vec{v}_1|^2 + U_1 = \frac{1}{2} m |\vec{v}_2|^2 + U_2$$

$$\Rightarrow \frac{1}{2} m (|\vec{v}_1|^2 - |\vec{v}_2|^2) = U_2 - U_1 \quad *$$

$$|\vec{p}_{\text{before}, \parallel}| = |\vec{p}_{\text{after}, \parallel}|$$

$$m |\vec{v}_1| \sin \theta_1 = m |\vec{v}_2| \sin \theta_2 \rightarrow \frac{|\vec{v}_1|}{|\vec{v}_2|} = \frac{\sin \theta_2}{\sin \theta_1}$$

$$* \Rightarrow \frac{1}{2} m |\vec{v}_1|^2 \left(1 - \frac{|\vec{v}_2|^2}{|\vec{v}_1|^2}\right) = U_2 - U_1$$

$$1 - \left(\frac{\sin \theta_1}{\sin \theta_2}\right)^2 = \frac{2}{m |\vec{v}_1|^2} (U_2 - U_1) \Rightarrow \frac{\sin \theta_1}{\sin \theta_2} = \sqrt{1 + \frac{2}{m |\vec{v}_1|^2} (U_2 - U_1)}$$

Problem

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$$\begin{aligned}
 \mathcal{L} &= \frac{1}{2} \sum_a m_a |\vec{v}_a|^2 - U : \vec{v}_a = \vec{v}_a' + \vec{V} \\
 &= \frac{1}{2} \sum_a m_a (\vec{v}_a' + \vec{V})^2 - U = \frac{1}{2} \sum_a m_a (\vec{v}_a'^2 + |\vec{V}|^2 + 2 \vec{v}_a' \cdot \vec{V}) - U \\
 &= \frac{1}{2} \sum_a m_a |\vec{v}_a'|^2 - U + \frac{1}{2} \sum_a m_a |\vec{V}|^2 + \sum_a m_a \vec{v}_a' \cdot \vec{V} \\
 &= \mathcal{L}' + \frac{1}{2} \mu |\vec{V}|^2 + \vec{V} \cdot \vec{P}'
 \end{aligned}$$

$$S = S' + \frac{1}{2} \mu |\vec{V}|^2 t + \vec{V} \cdot \sum_a m_a \vec{v}_a' \Rightarrow S = S' + \frac{1}{2} \mu |\vec{V}|^2 t + \mu \vec{V} \cdot \vec{R}'$$

Problem 1

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$$x_a = r_a \cos \phi_a \rightarrow \dot{x} = \dot{r} \cos \phi - r \dot{\phi} \sin \phi$$

$$y_a = r_a \sin \phi_a \rightarrow \dot{y} = \dot{r} \sin \phi + r \dot{\phi} \cos \phi$$

$$z_a = z_a \rightarrow \dot{z} = \dot{z}$$

$$\begin{aligned}
 M_x &= \sum_a m_a (y \dot{z} - z \dot{y}) = m [(r \sin \phi) \dot{z} - z (\dot{r} \sin \phi + r \dot{\phi} \cos \phi)] \\
 &= m (r \dot{z} - z \dot{r}) \sin \phi - m z r \dot{\phi} \cos \phi
 \end{aligned}$$

$$\begin{aligned}
 M_y &= \sum_a m_a (z \dot{x} - x \dot{z}) = m z (\dot{r} \cos \phi - r \dot{\phi} \sin \phi) - m r \dot{z} \cos \phi \\
 &= -m (r \dot{z} - z \dot{r}) \cos \phi - m z r \dot{\phi} \sin \phi
 \end{aligned}$$

$$M_z = [m(x\dot{y} - y\dot{x})] = m[(r\cos\phi)(\dot{r}\sin\phi + r\dot{\phi}\cos\phi) -$$

$$\hookrightarrow (r\sin\phi)(\dot{r}\cos\phi - r\dot{\phi}\sin\phi)] = \boxed{mr^2\dot{\phi}}$$

$$M^2 = M_x^2 + M_y^2 + M_z^2 = m^2(r\dot{z} - z\dot{r})^2 + m^2\dot{z}^2 r^2 \dot{\phi}^2 + m^2 r^4 \dot{\phi}^2$$

$$= \boxed{m^2(r\dot{z} - z\dot{r})^2 + m^2 r^2 \dot{\phi}^2 (z^2 + r^2)}$$

Problem 2

Page 21

$$x = r\sin\theta\cos\phi \rightarrow \dot{x} = \dot{r}\sin\theta\cos\phi + r\dot{\theta}\cos\theta\cos\phi - r\dot{\phi}\sin\theta\sin\phi$$

$$y = r\sin\theta\sin\phi \rightarrow \dot{y} = \dot{r}\sin\theta\sin\phi + r\dot{\theta}\cos\theta\sin\phi + r\dot{\phi}\sin\theta\cos\phi$$

$$z = r\cos\theta \rightarrow \dot{z} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$$

$$M_x = m(y\dot{z} - z\dot{y}) = m[r\sin\theta\sin\phi(\dot{r}\cos\theta - r\dot{\theta}\sin\theta) -$$

$$\hookrightarrow (r\cos\theta)(\dot{r}\sin\theta\sin\phi + r\dot{\theta}\cos\theta\sin\phi + r\dot{\phi}\sin\theta\cos\phi)]$$

$$= m[r\dot{r}\sin\theta\cancel{\cos\theta}\sin\phi - r^2\dot{\theta}\sin^2\theta\sin\phi - r\dot{r}\sin\theta\cancel{\cos\theta}\sin\phi]$$

$$\hookrightarrow -r^2\dot{\theta}\cos^2\theta\sin\phi - r^2\dot{\phi}\sin\theta\cos\theta\cos\phi]$$

$$= \boxed{-mr^2[\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi]}$$

$$M_y = m(z\dot{x} - x\dot{z}) = m[(r\cos\theta)(\dot{r}\sin\theta\cos\phi + r\dot{\theta}\cos\theta\cos\phi - r\dot{\phi}\sin\theta\sin\phi) -$$

$$\hookrightarrow -r\sin\theta\cos\phi(\dot{r}\cos\theta - r\dot{\theta}\sin\theta)]$$

$$= m[r\dot{r}\sin\theta\cos\theta\cos\phi + r^2\dot{\theta}\cos^2\theta\cos\phi - r^2\dot{\phi}\sin\theta\cos\theta\sin\phi - r\dot{r}\sin\theta\cancel{\cos\theta}\cos\phi]$$

$$\hookrightarrow +r^2\dot{\theta}\sin^2\theta\cos\phi]$$

$$= \boxed{mr^2[\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi]}$$

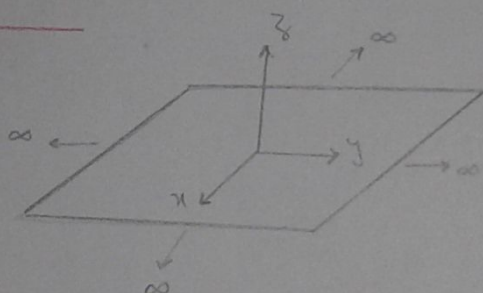
$$\begin{aligned}
 M_z &= m(x\dot{y} - y\dot{x}) = m[r\sin\theta\cos\phi(\dot{v}\sin\theta\sin\phi + r\dot{\theta}\cos\theta\sin\phi + \\
 &\quad \hookrightarrow v\dot{\phi}\sin\theta\cos\phi) - r\sin\theta\sin\phi(\dot{v}\sin\theta\cos\phi + v\dot{\theta}\cos\theta\cos\phi - r\dot{\phi}\sin\theta\sin\phi)] \\
 &= m[r\dot{v}\sin^2\theta\sin\phi\cos\phi + r^2\dot{\theta}\sin\theta\cos\theta\sin\phi\cos\phi + r^2\dot{\phi}\sin^2\theta\cos^2\phi \\
 &\quad \hookrightarrow -r\dot{v}\sin^2\theta\sin\phi\cos\phi - r^2\dot{\theta}\sin\theta\cos\theta\sin\phi\cos\phi + r^2\dot{\phi}\sin^2\theta\sin^2\phi] \\
 &= m r^2 \dot{\phi} \sin^2\theta
 \end{aligned}$$

$$\begin{aligned}
 M^2 &= M_x^2 + M_y^2 + M_z^2 = m^2 v^4 (\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi)^2 \\
 &\quad \hookrightarrow + m^2 v^4 (\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi)^2 + m^2 v^4 \dot{\phi}^2 \sin^4\theta \\
 &= m^2 v^4 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2\theta)
 \end{aligned}$$

Problem 3

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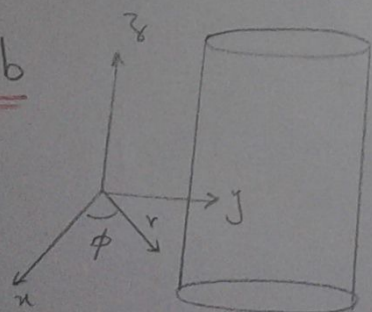
a



Conservative Quantities:

$\vec{P}_x$, $\vec{P}_y$, $\vec{M}_z$

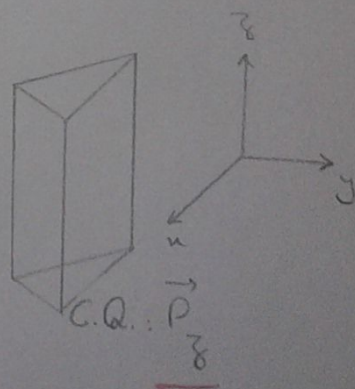
b



C. Q. :

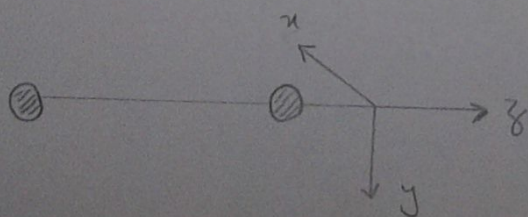
$\vec{P}_z$, $\vec{M}_z$

c



C. Q. : $\vec{P}_z$

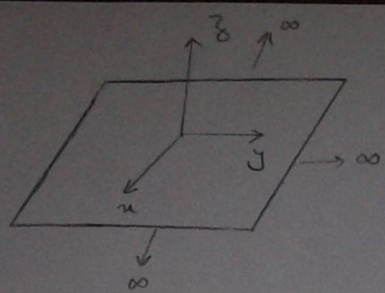
d



C. Q. :

$\vec{M}_z$

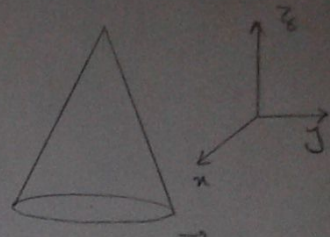
e



C. Q. :

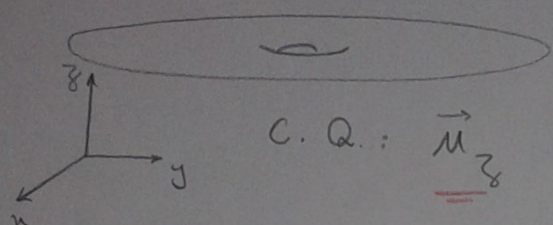
$$\underline{\underline{\vec{p}_x}}$$

f



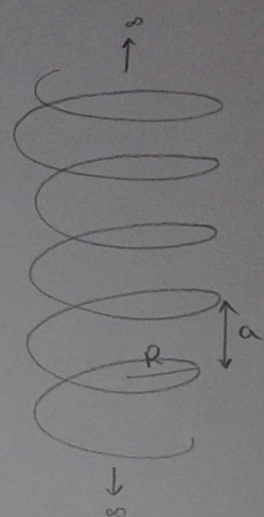
C. Q. : $\underline{\underline{\vec{M}_z}}$

g



C. Q. : $\underline{\underline{\vec{M}_z}}$

h



$$\frac{a}{2\pi} \delta \phi = \delta z$$

$$\text{if: } \delta \phi = 2\pi \Rightarrow \delta z = a$$

$$\delta \mathcal{L} = \delta z \frac{d\mathcal{L}}{dz} + \delta \phi \frac{d\mathcal{L}}{d\phi}$$

$$= \delta z \dot{p}_z + \delta \phi \dot{M}_z$$

$$= \delta \phi \left(\frac{a}{2\pi} \dot{p}_z + \dot{M}_z \right) = 0$$

$$\Rightarrow \frac{a}{2\pi} \dot{p}_z + \dot{M}_z = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{a}{2\pi} p_z + M_z \right) = 0$$

$$\text{Q.C. : } \underline{\underline{\frac{a}{2\pi} p_z + M_z}}$$

Problem 1

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$$\mathcal{L}' = \frac{1}{2} m' \vec{v}'^2 - U(v')$$

$$= \frac{\alpha^2}{\beta^2} \mu \frac{1}{2} m |\vec{v}|^2 - \alpha^k U(v) : \quad \frac{(\frac{L'}{L})}{(\frac{T'}{T})} = \frac{\alpha}{\beta}, \quad \frac{m'}{m} = \mu$$

$$\text{if: } \frac{\alpha^2}{\beta^2} \mu = \alpha^k \Rightarrow \beta = (\alpha^{1-\frac{k}{2}}) (\mu^{\frac{1}{2}})$$

$$\Rightarrow \frac{T'}{T} = \left(\frac{L'}{L}\right)^{1-\frac{k}{2}} \left(\frac{m'}{m}\right)^{\frac{1}{2}} \xrightarrow{L=L'} \frac{T'}{T} = \sqrt{\frac{m'}{m}}$$

Problem 2

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$$\left. \begin{aligned} L=L' &\Rightarrow \alpha = \frac{L'}{L} = 1 \\ m=m' &\Rightarrow \mu = \frac{m'}{m} = 1 \\ \frac{U'}{U} = 0, \frac{T'}{T} = \beta \end{aligned} \right\}$$

$$\mathcal{L}' = \frac{1}{2} m' \vec{v}'^2 - U' = \frac{1}{\beta^2} \frac{1}{2} m |\vec{v}|^2 - 0 U$$

$$\Rightarrow \frac{1}{\beta^2} = 0 \Rightarrow \beta^2 = \frac{1}{0} \Rightarrow \frac{T'}{T} = \sqrt{\frac{U}{U'}}$$

Problem 1

Page 26

$$\left. \begin{aligned} E &= \frac{1}{2} m \left(\frac{dx}{dt}\right)^2 - mgl \cos \phi : dx = L d\phi \\ E &= \frac{1}{2} m L^2 \dot{\phi}^2 - mgl \cos \phi = -mgl \cos \phi \end{aligned} \right\}$$

$$\begin{aligned} T &= 2\sqrt{2m} \int_{\phi_0}^{\phi_0} \frac{dx}{\sqrt{E - U(x)}} \\ &= 2\sqrt{2m} \int_{\phi_0}^{\phi_0} \frac{L d\phi}{\sqrt{mgl \cos \phi - mgl \cos \phi_0}} \end{aligned}$$

$$T = 2 \frac{\sqrt{2m}}{\sqrt{mgl}} L \int_0^{\phi_0} \frac{d\phi}{\sqrt{\cos\phi - \cos\phi_0}} = 2 \sqrt{\frac{2L}{g}} \int_0^{\phi_0} \frac{d\phi}{\sqrt{\cos\phi - \cos\phi_0}} ; \cos\phi = 1 - 2\sin^2 \frac{\phi}{2}$$

$$T = 2 \sqrt{\frac{2L}{g}} \int_0^{\phi_0} \frac{d\phi}{\sqrt{2\sin^2 \frac{\phi_0}{2} - 2\sin^2 \frac{\phi}{2}}} = 2 \sqrt{\frac{L}{g}} \int_0^{\phi_0} \frac{d\phi}{\sqrt{\sin^2 \frac{\phi_0}{2} - \sin^2 \frac{\phi}{2}}} \quad (*)$$

1. We replace $\frac{\sin \frac{\phi}{2}}{\sin \frac{\phi_0}{2}}$ by $\sin \xi$, then we have:

$$\frac{\sin \frac{\phi}{2}}{\sin \frac{\phi_0}{2}} = \sin \xi \Rightarrow \frac{1}{2} \frac{\cos \frac{\phi}{2}}{\sin \frac{\phi_0}{2}} d\phi = \cos \xi d\xi \Rightarrow d\phi = \frac{2 \sin \frac{\phi_0}{2} \cos \xi}{\cos \frac{\phi}{2}} d\xi$$

$$\Rightarrow d\phi = \frac{2 \sin \frac{\phi_0}{2} \sqrt{1 - \sin^2 \xi}}{\sqrt{1 - \sin^2 \frac{\phi}{2}}} d\xi \quad (**)$$

$$\frac{\sin \frac{\phi_0}{2}}{\sin \frac{\phi_0}{2}} = 1 = \sin \left(\frac{\pi}{2} \right)$$

2. And we put ** in * :

$$T = 2 \sqrt{\frac{L}{g}} \int_0^{\frac{\pi}{2}} \frac{2 \sin \frac{\phi_0}{2} \sqrt{1 - \sin^2 \xi} d\xi}{\sqrt{1 - \sin^2 \frac{\phi_0}{2} \sin^2 \xi} \left(\sin \frac{\phi_0}{2} \right)}$$

$$= 4 \sqrt{\frac{L}{g}} \int_0^{\frac{\pi}{2}} \frac{d\xi}{\sqrt{1 - \sin^2 \frac{\phi_0}{2} \sin^2 \xi}}$$

3. Last integral is a complete elliptic integral of the first kind, general solution is as follows:

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{d\xi}{\sqrt{1 - k^2 \sin^2 \xi}} = \frac{\pi}{2} \sum_{n=0}^{\infty} \left(\frac{(2n)!}{2^{2n} (n!)^2} \right)^2 k^{2n} = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \dots \right)$$

if: $\sin \frac{\phi_0}{2} \approx \frac{\phi_0}{2} \ll 1 \Rightarrow T = 4 \sqrt{\frac{L}{g}} \frac{\pi}{2} \left(1 + \frac{1}{4} \left(\frac{\phi_0}{2} \right)^2 + \dots \right) = 2\pi \sqrt{\frac{L}{g}} \left(1 + \frac{\phi_0^2}{16} + \dots \right)$

Problem 2

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$$\begin{aligned}
 \underline{b} \quad T &= 2\sqrt{2m} \int_0^{x_0} \frac{dx}{\sqrt{E_0 + \frac{U_0}{\cosh^2 \alpha x}}} = 2\sqrt{2m} \int_0^{x_0} \frac{dx \cosh \alpha x}{\sqrt{E_0 \cosh^2 \alpha x + U_0}} \quad : -\frac{U_0}{\cosh^2 \alpha x} = E_0 \\
 &\Rightarrow x_0 = \frac{1}{\alpha} \operatorname{arcCosh} \sqrt{\frac{U_0}{E_0}} \\
 &= 2\sqrt{2m} \int_0^{x_0} \frac{dx \cosh \alpha x}{\sqrt{E_0 + U_0 + E_0 \sinh^2 \alpha x}} \quad : \cosh^2 \alpha x = 1 + \sinh^2 \alpha x \\
 &= \frac{2\sqrt{2m}}{\sqrt{E_0}} \int_0^{x_0} \frac{dx \cosh \alpha x}{\sqrt{\frac{E_0 + U_0}{E_0} + \sinh^2 \alpha x}} \quad : U = \sinh \alpha x \Rightarrow du = \alpha \cosh \alpha x dx \\
 &\Rightarrow dx = \frac{1}{\alpha \cosh \alpha x} du \\
 &= \frac{2}{\alpha} \sqrt{\frac{2m}{E_0}} \int_0^{U_1} \frac{du}{\sqrt{k^2 + u^2}} \quad : x=0 \Rightarrow U = \sinh \alpha \cdot 0 = 0 \\
 &x = \frac{1}{\alpha} \operatorname{arcCosh} \sqrt{\frac{U_0}{E_0}} \Rightarrow U_1 = \sinh \left(\alpha \frac{1}{\alpha} \operatorname{arcCosh} \sqrt{\frac{U_0}{E_0}} \right) \\
 &= \frac{2}{\alpha} \sqrt{\frac{2m}{E_0}} \int_0^{U_1} \frac{du}{k \sqrt{1 + \left(\frac{u}{k}\right)^2}} \quad : \frac{u}{k} = \zeta \Rightarrow du = k d\zeta, U=0 \Rightarrow \zeta=0 \\
 &U=U_1 \Rightarrow \zeta = \zeta_0 = \frac{U_1}{k} \\
 &= \frac{2}{\alpha} \sqrt{\frac{2m}{E_0}} \int_0^{\zeta_0} \frac{d\zeta}{\sqrt{1 + \zeta^2}} = \frac{2}{\alpha} \sqrt{\frac{2m}{E_0}} \operatorname{arcSinh} \zeta \Big|_0^{\zeta_0} = \frac{2}{\alpha} \sqrt{\frac{2m}{E_0}} \operatorname{arcSinh} \zeta_0
 \end{aligned}$$

▮ We know that: $\cosh^2 y - \sinh^2 y = 1$, so $\sinh y = \sqrt{\cosh^2 y - 1}$,
 if we put $y = \operatorname{arcCosh} y$, then we can conclude: $\sinh(\operatorname{arcCosh} y) = \sqrt{y^2 - 1}$ *

$$T = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh} \xi = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh} \frac{U_+}{k} = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh} \left(\frac{1}{\sqrt{1 + \frac{U_+}{E_+}}} U_+ \right)$$

$$T = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh} \left(\frac{1}{\sqrt{1 + \frac{U_+}{E_+}}} \sinh \left(\operatorname{arccosh} \sqrt{1 + \frac{U_+}{E_+}} \right) \right)$$

According to *

$$T = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh} \left(\frac{1}{\sqrt{1 + \frac{U_+}{E_+}}} \sqrt{-\frac{U_+}{E_+} - 1} \right) = \frac{2}{\alpha} \sqrt{\frac{2m}{E_+}} \operatorname{arcsinh}(i) = \frac{i\pi}{\alpha} \sqrt{\frac{2m}{|E_+|}}$$

$$T = \frac{i\pi}{\alpha} \sqrt{\frac{2m}{|E_+|}} = \frac{\pi}{\alpha} \sqrt{\frac{2m}{|E_+|}}$$

C

$$T = 2\sqrt{2m} \int_{x_0}^{x_1} \frac{dx}{\sqrt{E_+ - U_+ \tanh^2 \alpha x}} \quad ; \quad E_+ = U_+ \tanh^2 \alpha x_0 \Rightarrow x_0 = \frac{1}{\alpha} \operatorname{arctan} \sqrt{\frac{E_+}{U_+}}$$

$$= 2\sqrt{2m} \int_{x_0}^{x_1} \frac{dx \cos \alpha x}{\sqrt{E_+ \csc^2 \alpha x - U_+ \sin^2 \alpha x}} = 2\sqrt{2m} \int_{x_0}^{x_1} \frac{dx \cos \alpha x}{\sqrt{E_+ - (E_+ + U_+) \sin^2 \alpha x}}$$

$$= 2\sqrt{2m} \int_{x_0}^{x_1} \frac{\cos \alpha x dx}{\sqrt{\frac{E_+}{E_+ + U_+} - \frac{\sin^2 \alpha x}{\frac{E_+ + U_+}{k^2}}}}$$

$$: U = i \sin \alpha x \Rightarrow dU = i \alpha \cos \alpha x dx \Rightarrow dx = \frac{1}{i \alpha \cos \alpha x} dU$$

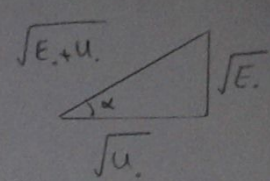
$$= \frac{2}{i \alpha} \sqrt{\frac{2m}{E_+ + U_+}} \int_{x_0}^{x_1} \frac{dU}{\sqrt{k^2 + U^2}}$$

$$x=0 \Rightarrow U = i \sin \alpha x = 0$$

$$x = \frac{1}{\alpha} \operatorname{arctan} \sqrt{\frac{E_+}{U_+}} \Rightarrow U = i \sin \left(\operatorname{arctan} \sqrt{\frac{E_+}{U_+}} \right)$$

$$\text{One can easily realize that : } x = \int \frac{dU}{\sqrt{k^2 + U^2}} = \operatorname{arcsin} \frac{U}{k} \Big|_{x_0}^{x_1}$$

$$T = \frac{2}{i\alpha} \sqrt{\frac{2m}{E_+ + U_+}} \operatorname{arcsinh} \frac{U_+}{k} \Big|_0^{U_+} = \frac{2}{i\alpha} \sqrt{\frac{2m}{E_+ + U_+}} \operatorname{arcsinh} \frac{U_+}{k}$$

$$\boxed{\operatorname{arctan} \sqrt{\frac{E_-}{U_+}} = \alpha \Rightarrow \sin(\operatorname{arctan} \sqrt{\frac{E_-}{U_+}}) = \frac{\sqrt{E_-}}{\sqrt{E_+ + U_+}}}$$


$$T = \frac{2}{i\alpha} \sqrt{\frac{2m}{E_+ + U_+}} \operatorname{arcsinh} \left(\frac{i}{\sqrt{\frac{E_-}{E_+ + U_+}}} \sin(\operatorname{arctan} \sqrt{\frac{E_-}{U_+}}) \right) = \frac{2}{i\alpha} \sqrt{\frac{2m}{E_+ + U_+}} \operatorname{arcsinh}(i)$$

$$T = \frac{2}{i\alpha} \sqrt{\frac{2m}{E_+ + U_+}} \frac{i\pi}{2} = \frac{\pi}{\alpha} \sqrt{\frac{2m}{E_+ + U_+}}$$

a

$$T = 2\sqrt{2m} \int_0^{x_+} \frac{dx}{\sqrt{E_+ - Ax^n}} \quad ; \quad E_+ = Ax_+^n \rightarrow x_+ = \left(\frac{E_+}{A}\right)^{\frac{1}{n}}$$

$$= 2\sqrt{2m} \int_0^{x_+} \frac{dx}{\sqrt{E_+} \sqrt{1 - \frac{A}{E_+} x^n}} \quad ; \quad \frac{A}{E_+} x^n = y^n \Rightarrow dx = \left(\frac{E_+}{A}\right)^{\frac{1}{n}} dy$$

$$x=0 \Rightarrow y=0$$

$$x = \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \Rightarrow y=1$$

$$= 2\sqrt{\frac{2m}{E_+}} \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \int_0^1 \frac{dy}{\sqrt{1-y^n}} \quad ; \quad y^n = u \Rightarrow dy = \frac{1}{n} u^{\frac{1}{n}-1} du$$

$$y=0 \Rightarrow u=0$$

$$y=1 \Rightarrow u=1$$

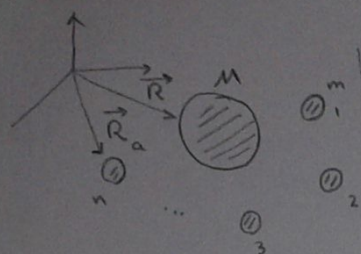
$$= \frac{2}{n} \sqrt{\frac{2m}{E_+}} \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \int_0^1 u^{\frac{1}{n}-1} (1-u)^{-\frac{1}{2}} du \quad ; \quad x \text{ is a Beta function}$$

$\boxed{\text{We know that: } \beta(p+1, q+1) = \int_0^1 u^p (1-u)^q du}$

$$= \frac{2}{n} \sqrt{\frac{2m}{E_+}} \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \beta\left(\frac{1}{n}, \frac{1}{2}\right) = \frac{2}{n} \sqrt{\frac{2m}{E_+}} \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \frac{\Gamma(\frac{1}{n}) \Gamma(\frac{1}{2})}{\Gamma(\frac{1}{n} + \frac{1}{2})} = \frac{2}{n} \sqrt{\frac{2m}{E_+}} \left(\frac{E_+}{A}\right)^{\frac{1}{n}} \frac{\Gamma(\frac{1}{n})}{\Gamma(\frac{1}{n} + \frac{1}{2})}$$

Problem

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$$\mathcal{L} = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} m [\dot{\vec{R}}_a]^2 - U$$

$$= \frac{1}{2} M \left(\frac{m}{M+nm} \right)^2 (\dot{\vec{r}}_a)^2 + \frac{1}{2} m [\dot{(\vec{r}_a + \vec{R})}]^2 - U$$

I

$$\left. \begin{aligned} \vec{r}_a &\equiv \vec{R}_a - \vec{R} \\ M \vec{R} + [m \vec{R}_a] &= 0 \end{aligned} \right\} \rightarrow \begin{aligned} M \vec{R} + [m (\vec{r}_a + \vec{R})] &= 0 \\ M \vec{R} + nm \vec{R} + [m \vec{r}_a] &= 0 \\ \vec{R} &= -\frac{m}{M+nm} [\vec{r}_a] \\ \dot{\vec{R}} &= -\frac{m}{M+nm} [\dot{\vec{r}}_a] \end{aligned}$$

$$I: \frac{1}{2} m [\dot{(\vec{r}_a + \vec{R})}]^2$$

$$= \frac{1}{2} m [(\dot{\vec{r}}_a + \dot{\vec{R}} + 2\dot{\vec{r}}_a \cdot \dot{\vec{R}})]$$

$$= \frac{1}{2} m [\dot{\vec{r}}_a^2 + \dot{\vec{R}}^2 + 2\dot{\vec{r}}_a \cdot \dot{\vec{R}}]$$

$$\hookrightarrow m [\dot{\vec{r}}_a \cdot \dot{\vec{R}}]$$

$$I: = \frac{1}{2} m [\dot{\vec{r}}_a^2 + \frac{1}{2} mn \dot{\vec{R}}^2 + m \dot{\vec{R}} \cdot [\dot{\vec{r}}_a]$$

$$* = \frac{1}{2} m [\dot{\vec{r}}_a^2 + \frac{1}{2} mn \left(\frac{m}{M+nm} \right)^2 (\dot{\vec{r}}_a)^2 + m \left(-\frac{m}{M+nm} \right) (\dot{\vec{r}}_a)^2]$$

$$\mathcal{L} = \frac{1}{2} \frac{Mm^2}{(M+nm)^2} (\dot{\vec{r}}_a)^2 + \frac{1}{2} m [\dot{\vec{r}}_a^2 + \frac{1}{2} \frac{nm^3}{(M+nm)^2} (\dot{\vec{r}}_a)^2 - \frac{m^2}{M+nm} (\dot{\vec{r}}_a)^2] - U$$

$$= \frac{1}{2} m [\dot{\vec{r}}_a^2 + (\dot{\vec{r}}_a)^2 \left[\frac{1}{2} \frac{Mm^2}{(M+nm)^2} + \frac{1}{2} \frac{nm^3}{(M+nm)^2} - \frac{m^2}{M+nm} \right] - U$$

$$= \frac{1}{2} m [\dot{\vec{r}}_a^2 + (\dot{\vec{r}}_a)^2 \left[\frac{Mm^2 + nm^3 - 2m^2(M+nm)}{2(M+nm)^2} \right] - U$$

$$= \frac{1}{2} m [\dot{\vec{r}}_a^2 - \frac{1}{2} \frac{m^2}{(M+nm)} (\dot{\vec{r}}_a)^2] - U$$